

LiteRehab Fusion

Real-time guidance for more consistent upper-limb rehabilitation practice at home.

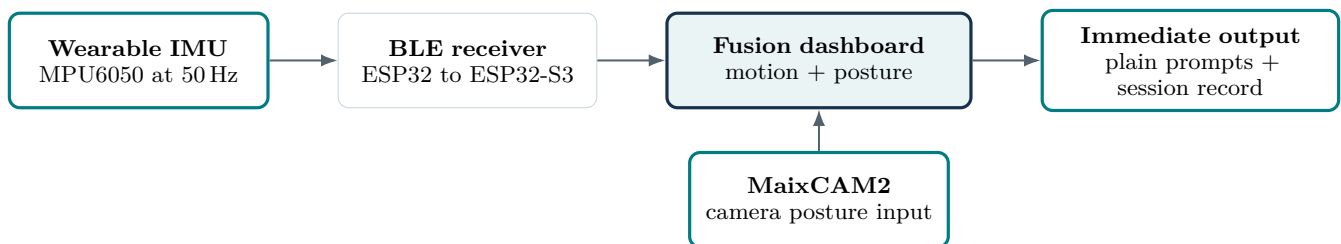
The gap

A patient can leave a physiotherapy appointment knowing which exercises to practise. At home, the harder question is whether each repetition is controlled, large enough, or being completed by leaning the trunk. Useful feedback may not arrive until the next appointment, after the practice has already happened.

Our response

LiteRehab Fusion gives feedback while the exercise is happening. The wearable and camera-based prototype identifies the demonstrated movement, counts repetitions, flags common practice errors, and records the session for later review.

How it works



What the MVP demonstrates

- Recognises the demonstrated elbow flexion and forearm rotation exercises.
- Gives immediate feedback for excessive speed, insufficient range, and trunk compensation.
- Shows repetition count on the wearable, with local LED and buzzer feedback.
- Keeps a synchronized record of IMU and pose information.
- Runs a classroom CNN-BiGRU baseline with a rule-based fallback.
- Passes **70 Python tests**, **three C host tests**, a model-loading smoke test, and builds for both ESP32 targets.

Route to use

The proposed first step is a supervised pilot with physiotherapists. A later version could be supplied through rehabilitation providers as part of a prescribed home-exercise programme. The physiotherapist would choose the exercises and review the records.

Who benefits

PATIENT

A clear cue arrives during the repetition instead of days later.

PHYSIOTHERAPIST

A session record could show what happened between appointments, while the physiotherapist keeps responsibility for exercise selection and clinical decisions.

Current limits

LiteRehab Fusion is an engineering prototype, **not a medical device**. It does not diagnose, prescribe treatment, score recovery, or replace professional supervision. The current model uses a small public dataset, supports a limited exercise set, and makes no clinical accuracy claim.

What we need next

We are looking for physiotherapist partners to test usability, refine the feedback, and collect representative movement data under professional supervision.